Liquids & Solids

Chapter 9

Chemistry Principles and Reactions

Gases vs. Liquids & Solids

- Gases follow ideal gas law, “under ordinary T & P”
- However, there is less “law and order” for Liquids and Solids
- Topics for this chapter:
  - Phase Diagrams
  - Intermolecular forces

Comparison of the Three Phases of Water

Phase	Molecular Spacing	Density	Volume of One Mole
Steam ($H_2O(g)$)		at $100^\circ C$, 1 atm 0.00059 g/mL	at $100^\circ C$, 1 atm 31 L
Water ($H_2O(l)$)		at $100^\circ C$, 1 atm 0.95 g/mL	19 mL
Ice ($H_2O(s)$)		at $0^\circ C$, 1 atm 0.92 g/mL	20 mL

9.1 Gases vs. Liquids & Solids

- Gaseous molecules are much **further apart**

- **Intermolecular forces** play a **negligible** role for gases

- Opposite statements hold for liquids & solids
  - Tightly packed & intermolecular forces give rise to important physical properties.

9.2 Liquid-Vapor Equilibrium

- Vaporization – process of a liquid turning into a gas (vapor)
  - Under the right conditions, in an open container, vaporization will continue until no liquid remains
  - In a **closed container**, **equilibrium** is established:
    - Rate of vaporization = rate of condensation (reverse process, g → l)
    - Liquid ⇌ vapor

9.2 **Vapor Pressure** inLiquid-Vapor Equilibrium

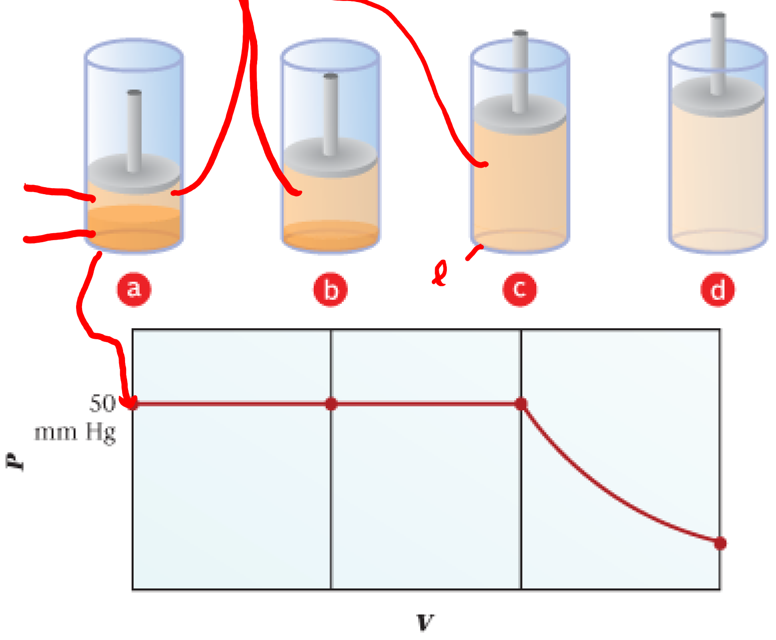
- Once equilibrium is established,
  - # of molecules of gas/unit volume (n/V = constant) is constant, thus pressure exerted by gas is also constant
  - Vapor P = nRT/V = constant ***if liquid is present***

    $R \cdot T \times \frac{1}{V}$

$\frac{n}{V}$ = constant

g

l



- If <u>both liquid & gas are present</u>, the vapor pressure of the gas <u>does not depend on volume</u>$_{\text{container}}$
  - Dark orange = liquid

- If liquid is not present, PV = nRT holds
  - This is technically Boyle's law ($P_1 V_1 = P_2 V_2$)

Vapor Pressure vs. Temperature

- Vapor Pressure & Temperature are directly proportional
  - If T goes up, vapor pressure goes up.
  - Easier/quicker for H_2O to evaporate in Tucson, Az than here in Groton, CT

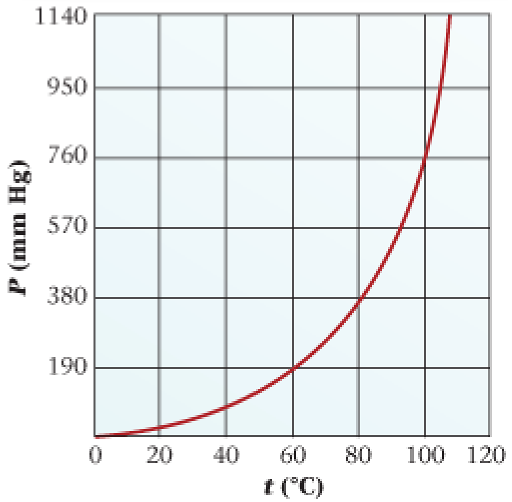
(This is actually near Yuma, Az)

Vapor Pressure vs. Temperature

- Vapor Pressure & Temperature are directly proportional (left figure)

A plot of vapor pressure vs. temperature (in °C) for a liquid is an exponentially increasing curve.

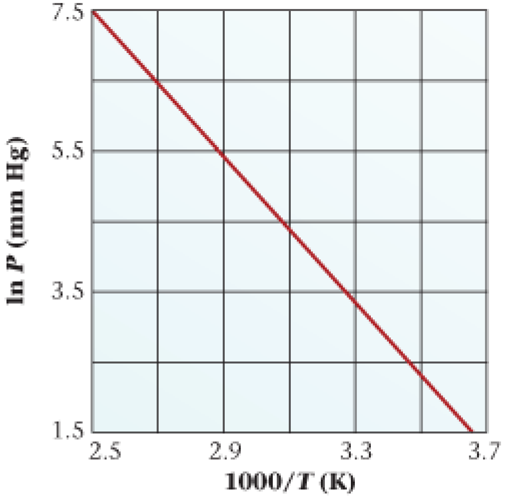
Vapor Pressure



Temp

A plot of the logarithm of the vapor pressure vs. 1/*T* (in K) is a straight line. (To provide larger numbers, we have plotted 1000/*T*.)

ln P



$\frac{1}{T}$

y=mx+b

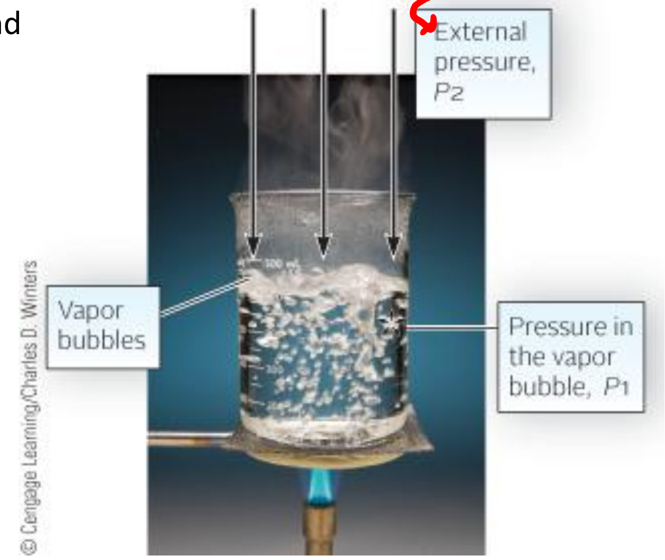
- Right figure: easier to understand b/c it's linear
- $y - y_1 = m\,(x - x_1)$ point slope
- $\ln P_1 - \ln P_2 = -(\Delta H_{vap}/R)\,(1/T_1 - 1/T_2)$
  - **Clausius-Clapeyron Equation**
  - R = 8.31 J/(mol*K) & ΔH_{vap} in J (Joules)

Boiling Point (B.P)

- B.P. = the temperature at which liquid boils into gas.
  - For a pure liquid, this temperature remains constant throughout the boiling process.
    - A pure liquid is a **single species:** an element (Hg), or an ionic (NaCl) or molecular (H_2O) compound
- B.P. depends on pressure above the liquid (P_2).
  - Vapor pressure of the liquid must ≥ P_2 for vapor bubbles to develop.
    - If P_2 = 1 atm (760 mm Hg), this is the <u>normal boiling point</u>.
      - Water's normal boiling point is 100 °C (at 1 atm).
    - Lowering P_2 can allow liquids to boil at lower temperatures.
      - B.P. of H_2O in Salt Lake City Utah is 95 °C

-higher altitude, lower P_2 (Barometric pressure)

Ambient,



Critical Temperature & Pressure

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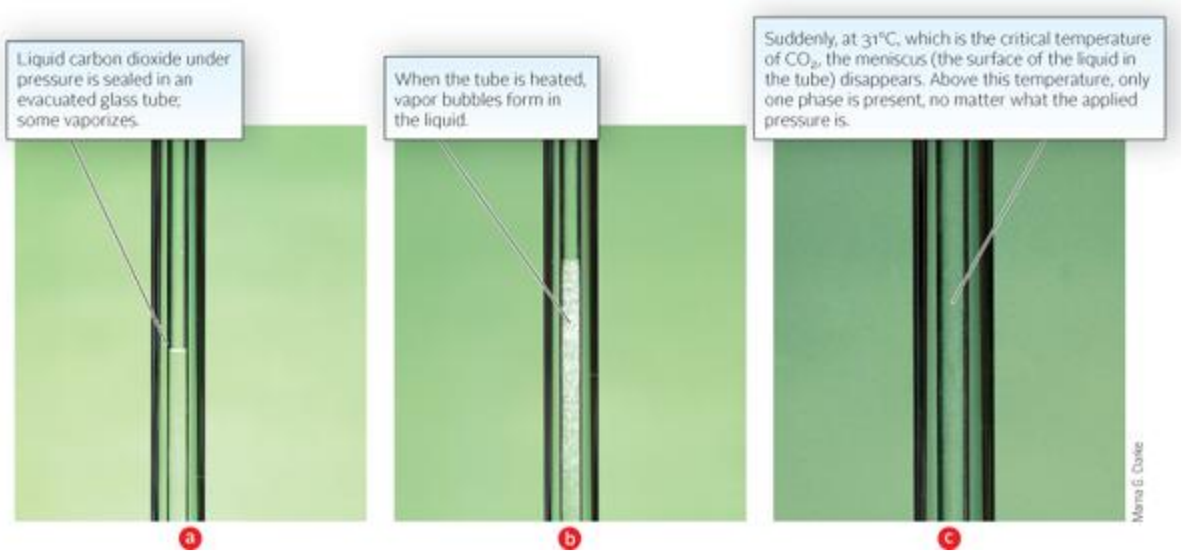
- Critical Temperature (C.T.) = temperature above which liquid phase of a pure substance cannot exist
  - Critical Pressure: pressure to condense gas → liquid at C.T.
    - Vapor pressure of liquid at C.T.
  - Above C.T. & C.P. the substance becomes a ***supercritical fluid***

0 °C: 34 atm	liquid & gas
10 °C: 44 atm	less liquid & gas
20 °C: 56 atm	even less l & g
31 °C: 73 atm	no liquid, only gas



Critical temperatures of various species

- Permanent gases: require cooling and pressure to condense from gas → liquid

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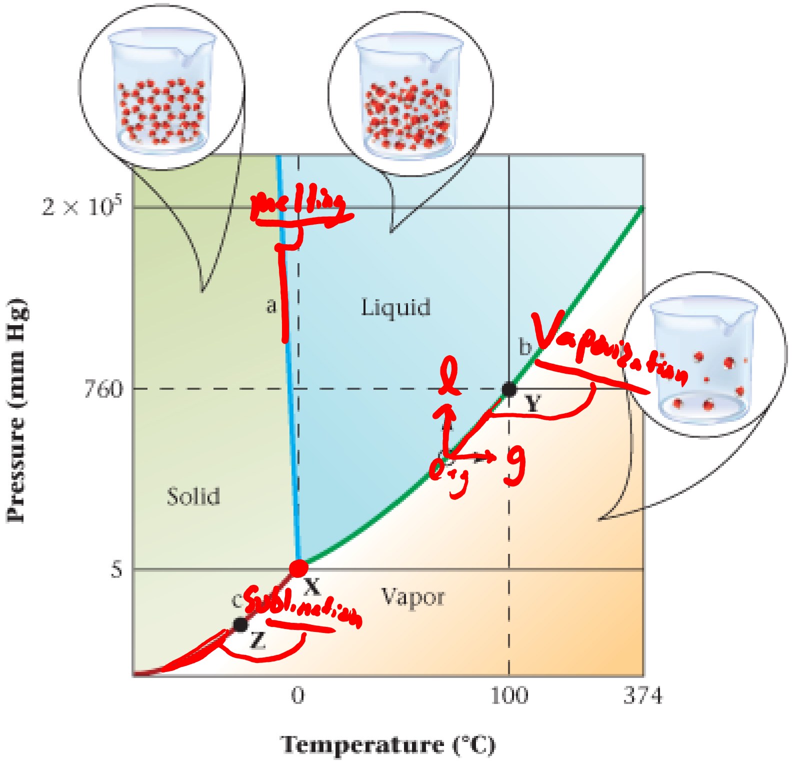
Permanent Gases		Condensable Gases		Liquids	
Helium	−268	Carbon dioxide	31	Ethyl ether	194
Hydrogen	−240	Ethane	32	Ethyl alcohol	243
Nitrogen	−147	Propane	97	Benzene	289
Argon	−122	Ammonia	132	Bromine	311
Oxygen	−119	Chlorine	144	Water	374
Methane	−82	Sulfur dioxide	158		

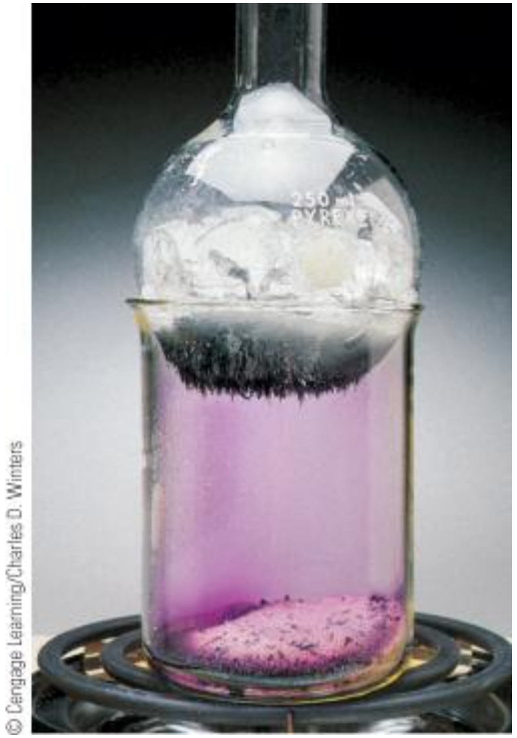
9.3 Phase Diagrams

- Phase diagram: A graph that shows the T & P at which different phases are in equilibrium with each other.

  - Along curves between zones, two phases are present
  - w/in zones, single phase present
    - Consider open circle and directions of change
  - Point **X**: **Triple Point**: where all three phases are present



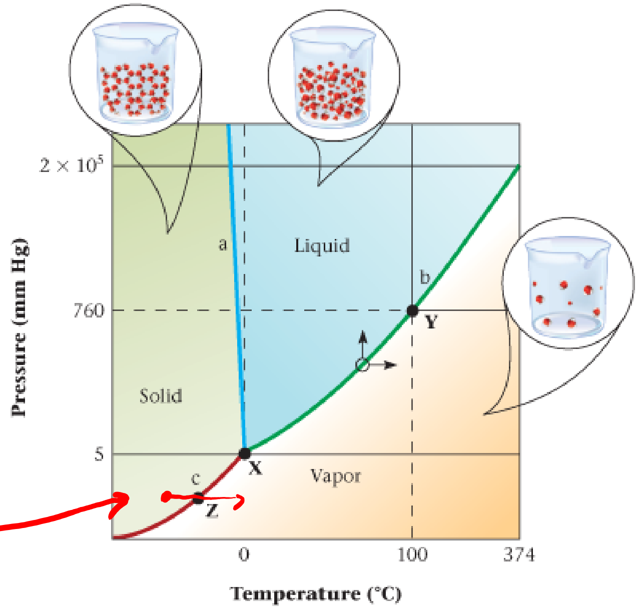
Sublimation (curve c, red)



- **Sublimation**: process by which a solid changes directly to a gas
  - $I_2\,(s) \rightleftharpoons I_2\,(g)$ →
  - Opposite of Sublimation: **Deposition**

- Freeze dried food →

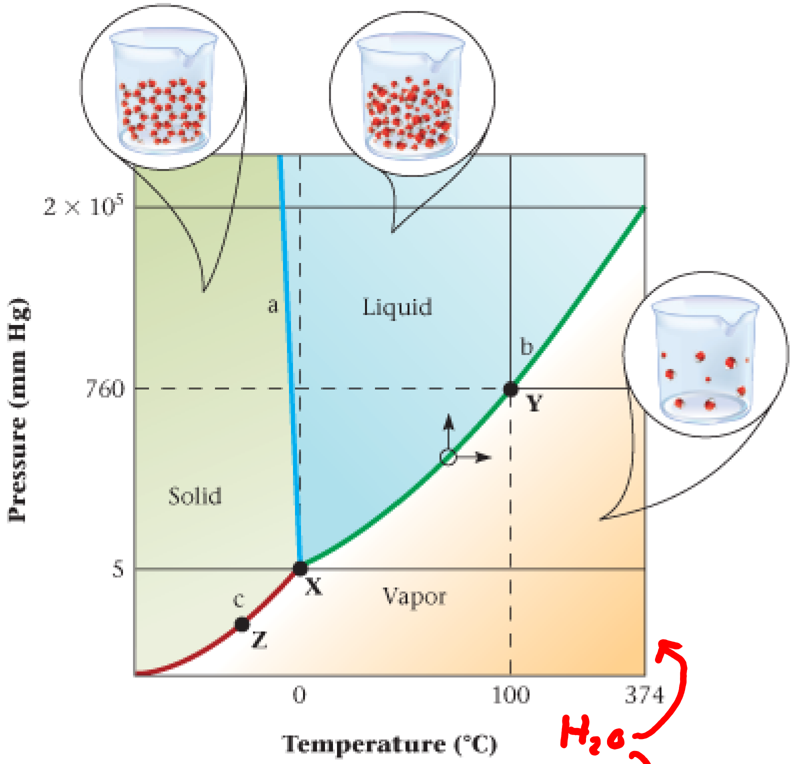
Keeping P fixed & slowly raising T removes $H_2O(g)$



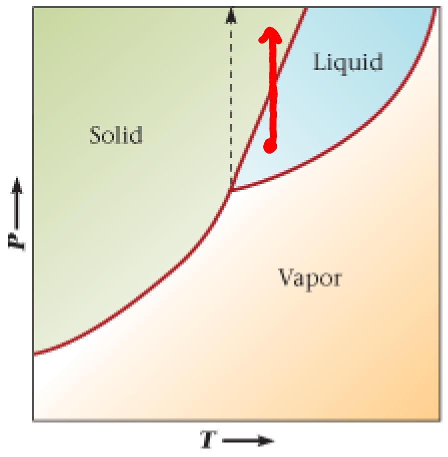
Melting Point: solid → liquid

- Normal melting point at 1 atm (760 mm Hg) pressure
  - Notice there is a very small difference between normal melting point & triple point (**X**) Temps

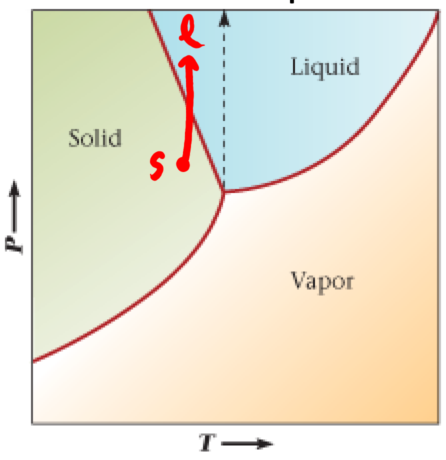
- *An increase in pressure favors the denser phase*
  - Below Left: denser solid
  - Below Right: denser liquid



denser solid



denser liquid



9.4 Molecular Compounds & Their Intermolecular Forces

- Molecular compounds (those w/ covalent bonds) are
  - Generally poor conductors of electricity due to being uncharged
    - Example: CH_4 Exception: $HCl\ (g) \rightarrow H^+\ (aq) + Cl^-\ (aq)$
  - Generally insoluble in H_2O but soluble in organic nonpolar solvents ($CHCl_3$, benzene, diethyl ether)
    - Exceptions: ethanol, ammonia (due to strong intermolecular forces)
  - Low melting & boiling point
    - Many molecules are gases under ambient conditions (N_2, O_2, CO_2)
    - Due to generally weak intermolecular forces in molecules

Key

  - **Stronger intermolecular forces lead to higher B.P./M.P.**

Intermolecular forces vs. covalent bonds

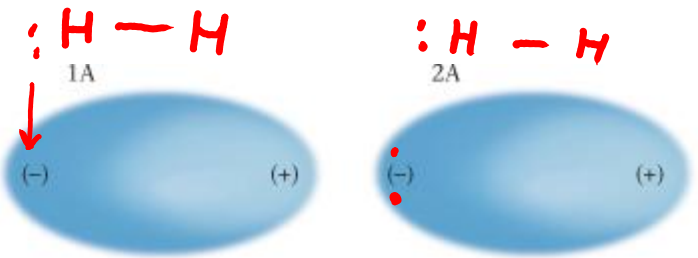
- Intercollegiate vs intramural sports
- Between school's vs within a school

- Intermolecular forces vs intramolecular (covalent bonds)
- Between molecules vs within a molecule

Intermolecular forces

- **Stronger intermolecular forces lead to higher B.P./M.P.**
- 3 types
    - London dispersion forces (weakest)
    - Dipole forces
    - Hydrogen bonds (strongest)

London Dispersion Forces



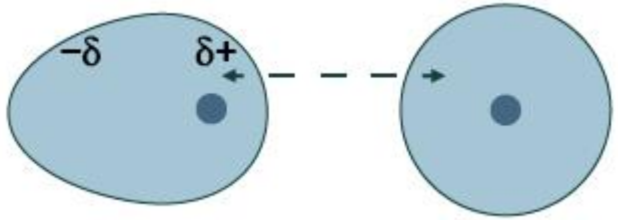
– Major force in Nonpolar molecules CO_2, CCl_4, CH_4, SF_6

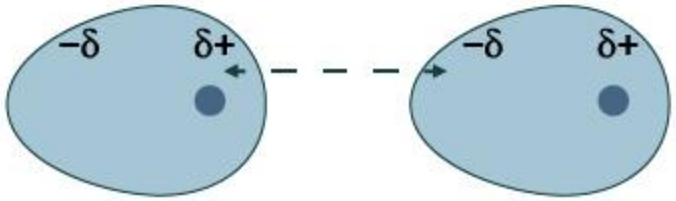
- Arise from an attraction between temporary or *induced* dipoles in adjacent molecules
- For H_2, on average the e^- are similarly close to both nuclei,
  - But, at a certain moment in time, the electrons might be closer to one nuclei than the other, creating a temporary dipole (+ & – in 1A).
    - This can induce a dipole in another molecule of H_2 (+ & – in 2A).
- Dispersion forces increase w/ increasing molar mass
  - C_4H_{10} has more dispersion forces than CH_4

London Dispersion Forces
For a Nobel Gas (i.e. 2 helium atoms)

Helium atoms
-no permanent dipole

- chance leads the left atom's e^- to be on the left, developing a partial negative charge (δ–)

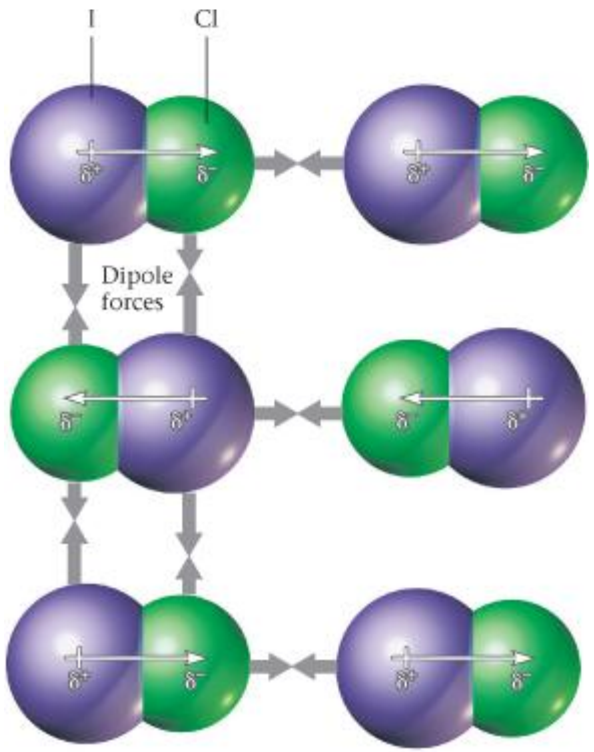




- This *temporary* dipole in the left atom *induces* a dipole in the right atom

Dipole Forces

- Polar molecules can develop temporary dipoles and thus exhibit London dispersion forces,
  - Important but off topic: CCl_4 is a nonpolar molecule with polar bonds!
- But, more importantly they exhibit dipole forces due to their permanent dipole
  - Molecules line up so (–) end of one molecule is close to (+) end of another
  - Generally, Dipole Forces > London Dispersion Forces



Boiling Points of Nonpolar versus Polar Substances

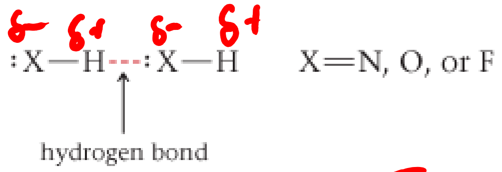
Nonpolar			Polar		
Formula	MM (g/mol)	bp (°C)	Formula	MM (g/mol)	bp (°C)
N_2	28	−196	CO	28	−192
SiH_4	32	−112	PH_3	34	−88
GeH_4	77	−90	AsH_3	78	−62
Br_2	160	59	ICl	162	97

Hydrogen Bonds

- Occur in molecules containing an H–N, H–O, H–F bond
  - H bound to a small electronegative atom (in blue)
- Arises due to large $\Delta_{electronegativity}$ & small atomic radii of H, N, O, F

Effect of Hydrogen Bonding on Boiling Point

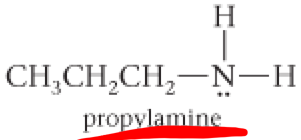
	bp (°C)		bp (°C)		bp (°C)
NH_3	−33	H_2O	100	HF	19
PH_3	−88	H_2S	−60	HCl	−85
AsH_3	−63	H_2Se	−42	HBr	−67
SbH_3	−18	H_2Te	−2	HI	−35

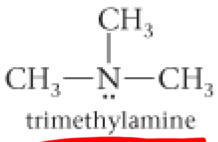


No H-bond
$H_3C-\ddot{O}-CH_3$

H-bond
CH_3CH_2-OH

MM (g/mol) 59 | 59 | 58





$CH_3CH_2CH_2CH_3$

B.P. (primary intermolecular force
49 °C (hydrogen bonds)
3 °C (dipole forces)
-1 °C (London disp. Force)

- Covalent bonds are stronger than intermolecular forces

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